Practice workbook · Evaluate kidney transplants and dialysis as treatments

Advantages and disadvantages of kidney transplant and dialysis, and why kidney donation is more common than other organs.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-TRANSPLANT-VS-DIALYSIS

1. Question 8 · part whole · 5 marks

June 2022 Paper 1R

Worked-example retry · Attempt part whole only.

8 The kidneys play an important role in the excretion of toxic waste.

Some people have kidney disease which reduces the effectiveness of the kidneys.

Kidney disease can be treated by dialysis or by a transplant.

Discuss the advantages and disadvantages of each of these treatments.

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(Total for Question 8 = 5 marks)

1. Question 8 · part whole · 5 marks · Working and answer

Working and answer

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2. Question 4 · part b · 5 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

4 The blood is filtered by the kidneys.

The filtrate formed passes through the kidney tubules and is changed into urine.

(a) The table lists the masses of different substances in blood, filtrate and urine passing through the kidneys in one day.

	Mass of substance in g per day		
Substance	**in blood**	**in filtrate**	**in urine**
protein	8000	0	0
glucose	180	180	0
urea	53	53	25
uric acid	9	9	1

Use the information from the table to answer these questions.

(i) Which substance does not pass into the filtrate? **(1)**

- ☒ **A** protein
- ☒ **B** glucose
- ☒ **C** urea
- ☒ **D** uric acid

(ii) The molecules of this substance do not pass into the filtrate.

What is the reason for this? **(1)**

- ☒ **A** the molecules are too large
- ☒ **B** the molecules are reabsorbed
- ☒ **C** the molecules are a waste product
- ☒ **D** the concentration of the molecules in the blood is too low

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2. Question 4 · part b · 5 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

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(iii) A person has diabetes but is not receiving treatment.

Explain why the mass of glucose per day in the urine would be different for this person.

(3)

(iv) Calculate the mass of urea in the urine as a percentage of the mass of urea in the filtrate.

(2)

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percentage = ... %

2. Question 4 · part b · 5 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

(b) Kidney failure can be treated by a kidney transplant or by dialysis.

Discuss the advantages and disadvantages of kidney transplants.

(5)

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(Total for Question 4 = 12 marks)

2. Question 4 · part b · 5 marks · Working and answer

Working and answer

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3. Question 7 · part b.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.i only.

7 People with polycystic kidney disease have cysts on their kidneys.

Cysts are sacs of fluid and they affect the functioning of the kidney.

Polycystic kidney disease is caused by a mutation of a gene.

(a) Explain how a gene mutation can affect the phenotype of an individual.

(3)

(b) Polycystic kidney disease can cause kidney failure.

Kidney failure can be treated using dialysis or a transplant.

(i) Describe the benefits of a transplant rather than dialysis as a treatment for kidney failure.

(2)

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3. Question 7 · part b.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.i only.

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(ii) In one year in the United Kingdom there were 4733 organ donations.

Of these donations, 3233 were kidney donations.

Calculate the percentage of organ donations that are kidney donations.

(2)

percentage = ... %

(iii) The number of kidney donations is much higher than the number of other organ donations such as heart or liver.

Give two reasons for the higher number of kidney donations.

(2)

1 .....

.....

2 .....

.....

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3. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

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4. Question 7 · part b.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

7 People with polycystic kidney disease have cysts on their kidneys.

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Polycystic kidney disease is caused by a mutation of a gene.

(a) Explain how a gene mutation can affect the phenotype of an individual.

(3)

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Kidney failure can be treated using dialysis or a transplant.

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(2)

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4. Question 7 · part b.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

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(ii) In one year in the United Kingdom there were 4733 organ donations.

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Give two reasons for the higher number of kidney donations.

(2)

1 .....

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2 .....

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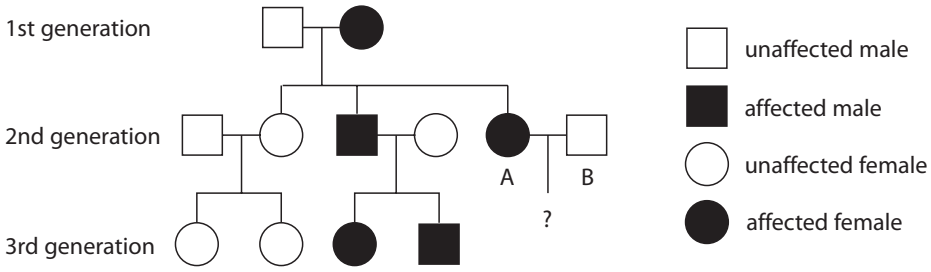
4. Question 7 · part b.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

(c) The diagram shows a family pedigree. Some of the family have polycystic kidney disease.

This disease is caused by a dominant allele that is not sex-linked.



(i) Determine the ratio of the females shown in the 3rd generation that are affected with polycystic kidney disease.

(1)

ratio = ...

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4. Question 7 · part b.iii · 2 marks · Working and answer

Working and answer

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5. Question 3 · part c · 2 marks

June 2024 Paper 2R

Further practice · Attempt part c only.

3 (a) Polycystic kidney disease (PKD) is an inherited condition caused by a gene mutation.

PKD affects the function of the kidneys.

State what is meant by a gene mutation.

(1)

.....

.....

(b) PKD occurs when a child inherits a faulty allele from one parent.

One parent has a genotype Hh and the other parent has a genotype hh.

The parents have four children.

Draw a genetic diagram to show the possible genotypes of the four children.

(2)

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(c) Some people with PKD need a kidney transplant.

Give one advantage and one disadvantage of a kidney transplant.

(2)

Advantage

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Disadvantage

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(Total for Question 3 = 5 marks)

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5. Question 3 · part c · 2 marks · Working and answer

Working and answer

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